

10th International Conference of Information and Communication Technology (ICICT-2020)

Identification of continuous stirred tank reactor based on PCA-interval type-2 fuzzy logic system method

Zhengmei Zhao*, Jun Li

School of Automation & Electrical Engineering, Lanzhou Jiaotong University, Lanzhou Gansu, China

Abstract

For unknown model of Continuous Stirred Tank Reactor (CSTR), a model identification based on the combination of principal component analysis (PCA) and a type-2 non-singular interval type-2 fuzzy logic system (Type-2 NSFLS-2) is proposed method. The PCA method can reduce the dimensionality of the high-dimensional input of the model. On this basis, type-2 NSFLS-2 identification model is established, and the backpropagation (BP) algorithm is used to update the parameters of antecedent and consequent membership function respectively. The method in this paper is applied to the example of CSTR, and it is compared with other types of FLS method and support vector machine (SVM) under the same conditions. The experimental results show that different types of interval type-2 FLS methods can achieve higher identification accuracy, among which Type-2 NSFLS-2 has higher identification accuracy.

© 2021 The Authors. Published by Elsevier B.V.

This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>)

Peer-review under responsibility of the scientific committee of the 10th International Conference of Information and Communication Technology.

Keywords: Interval type-2 fuzzy logic systems; principal component analysis; continuous stirred tank reactor; system identification

1. Introduction

The CSTR chemical process with strong nonlinear, time-varying, serious lag and complexity of external interference make model identification difficulty [1-4]. At present, neural network [5-7] and support vector machine (SVM) [8] have been widely used in chemical process identification. These successful application results show that the chemical process modeling using artificial intelligence methods such as neural network and SVM can overcome nonlinear problem of identification process. However, for complex chemical process, when the model needs expert

* Corresponding E-mail address: 654346113@qq.com

knowledge or contains uncertain information, these methods are often not enough to establish a reliable model, which may lead to the decline of identification performance index or poor generalization of the model.

As a powerful modeling tool, type-2 FLS method [9-10] has been successfully applied to power load forecasting [11], ladle furnace temperature prediction [12], traffic signal control [13], and has good application potential. Type-1 or type-2 FLS method is applied to the prediction modeling of complex nonlinear systems. When the input dimension is high, type-1 or type-2 FLS will encounter the rule explosion problem. Principal components analysis (PCA) is a better preprocessing method for feature extraction. In reference [14], a function connected neural network based on PCA was proposed, and the proposed method was successfully applied to the modeling of PTA production process. Considering the combination of PCA and interval type II FLS method, the rule "explosion" can be solved.

Therefore, on the basis of the successful application of the existing interval type-2 FLS method to the prediction modeling, considering the potential uncertainty factors of complex chemical process identification modeling, this paper proposes a method combining PCA with type-2 non-single value interval type-2 FLS, and further updates the parameters based on BP algorithm in the design process of interval type-2 FLS method. The proposed method is applied to a CSTR chemical process, and compared with SVM and other FLS methods under the same conditions to verify the effectiveness of the proposed method.

2. PCA algorithm

The PCA algorithm can eliminate the multicollinearity between variables and effectively reduce the noise and redundancy in the data. Dimensionality reduction technique by PCA algorithm can be considered for input data. Under the premise of ensuring accuracy, the original multi-dimensional variable data can be reduced in dimension, which will be beneficial to modeling of the type-2 FLS.

The PCA algorithm is implemented as follows:

Step1 given n-dimensional input, and data sets of N :

$$\tilde{X} = \{\tilde{x}_1, \dots, \tilde{x}_N\} \in \mathbf{R}^{n \times N}$$

And calculated mean of given data set.

Step2 calculated the covariance matrix for data set.

And calculated eigenvalues and eigenvectors of the covariance matrix

$$[U, A] = eig(NC) \quad (1)$$

where is the eigenvalue decomposition function.

Step3 projected the original training data into the subspace spanned by the set of p eigenvectors, which include the first p eigenvectors for covariance matrix of the training set,

$$x_i = U_p' \tilde{x}_i, i=1, \dots, N \quad (2)$$

Hence, the preprocessed data sets extracted the first p principal components using the PCA algorithm is as follows:

$$X = \{x_1, \dots, x_N\} \in \mathbf{R}^{p \times N} \quad (3)$$

For chemical processes, PCA technology is used to reduce the dimensionality of process variables, which can reflect a large number of process variables contained information about the operating conditions of the process, that is, it uses fewer irrelevant components to carry enough information.

3. Interval type-2 FLS

Suppose a series of N input-output data pairs are given, i.e.:

$$\left(\mathbf{x}^{(1)} : y^{(1)} \right), \dots, \left(\mathbf{x}^{(N)} : y^{(N)} \right) = \left\{ \mathbf{x}_1^{(t)}, \dots, \mathbf{x}_p^{(t)} : y^{(t)} \right\}_{t=1}^N \quad (4)$$

where $x_1 \in X_1, \dots, x_p \in X_p$ is the input processed by PCA algorithm, output is $y \in Y$. An interval type-2 FLS with M rules can be designed. Each rule has p antecedents, then i th rule R^i has:

$$R^i: \text{If } x_1 \text{ is } \tilde{F}_1^i, \dots, x_p \text{ is } \tilde{F}_p^i, \text{ then } y \text{ is } \tilde{G}^i, i=1, \dots, M. \quad (5)$$

where $\tilde{F}_k^i$ is the interval type-2 antecedent fuzzy sets corresponding to elements for $x^{(t)}$.

$\tilde{F}_k^i$ is described by the k th priori Gaussian principal membership function with uncertain mean, i.e.:

$$\mu_{\tilde{F}_k^i}(x_k) = \exp \left\{ -\frac{1}{2} \left(\frac{x_k - m_{\tilde{F}_k^i}}{\sigma_{\tilde{F}_k^i}} \right)^2 \right\}, m_{\tilde{F}_k^i} \in [m_{\tilde{F}_{k1}^i}, m_{\tilde{F}_{k2}^i}] \quad (6)$$

where, $k=1, \dots, p$, $i=1, \dots, M$.

In addition, all the points $x_k \in X_k$ are modeled as a type of Gaussian fuzzy number:

$$\mu_{x_k}(x_k) = \exp \left\{ -\frac{1}{2} \left(\frac{x_k - x_k'}{\sigma_{x_k}} \right)^2 \right\} \quad (7)$$

where, x_k' is center of fuzzy sets; σ_{x_k} is standard deviation of input set.

According to theorem 1 in the reference [15], when the input is a type of non-singular value, only fire set is affected. No matter what kind of t-norm is used, the operation result of input and antecedent is a fire interval.

After fuzzy inference, "center of set" is selected to reduce type, i.e.:

$$Y_{\cos}(Y^1, \dots, Y^M, F^1, \dots, F^M) = [y_l, y_r] = \int_{y^1} \dots \int_{y^M} \int_{f^1} \dots \int_{f^M} 1 \left/ \frac{\sum_{i=1}^M f^i y^i}{\sum_{i=1}^M f^i} \right. \quad (8)$$

where $Y_{\cos}$ is an interval set determined by two endpoints y_l and y_r ; $f^i \in F^i = [\underline{f}^i, \bar{f}^i]$; $y^i \in Y^i = [y_l^i, y_r^i]$, $i=1, \dots, M$.

4. BP algorithm design of interval type-2 FLS

An input-output data pair $(x_1^{(t)}, \dots, x_p^{(t)} : y^{(t)})$, $t=1, \dots, N$ is given, the objective is to design an interval type-2 FLS, then to minimize error function of formula (9) in a training epoch by updating the design parameters.

$$e^{(t)} = \frac{1}{2} \left[f(\mathbf{x}^{(t)}) - y^{(t)} \right]^2 \quad (9)$$

The BP algorithm is implemented as follows:

Step1 Initialize all antecedents, consequent and input parameters in membership function.

Step2 Set the counter $count=1$ for the training epoch.

Step3 Set the counter $t=1$ for the training samples.

Step4 Calculated y_l and y_r by K - M iterative method, and y_l and y_r are expressed as follows:

$$y_l = \frac{\sum_{i=1}^M f_l^i y_l^i}{\sum_{i=1}^M f_l^i} = \frac{\sum_{i=1}^L \bar{f}^i y_l^i + \sum_{j=L+1}^M \bar{f}^j y_l^j}{\sum_{i=1}^L \bar{f}^i + \sum_{j=L+1}^M \bar{f}^j} = y_l(\bar{f}^1, \dots, \bar{f}^L, \bar{f}^{L+1}, \dots, \bar{f}^M, y_l^1, \dots, y_l^M) \quad (10)$$

$$y_r = \frac{\sum_{i=1}^M f_r^i y_r^i}{\sum_{i=1}^M f_r^i} = \frac{\sum_{i=1}^R \bar{f}^i y_r^i + \sum_{j=R+1}^M \bar{f}^j y_r^j}{\sum_{i=1}^R \bar{f}^i + \sum_{j=R+1}^M \bar{f}^j} = y_r(\bar{f}^1, \dots, \bar{f}^R, \bar{f}^{R+1}, \dots, \bar{f}^M, y_r^1, \dots, y_r^M) \quad (11)$$

Step5 Calculated defuzzified output of interval type-2 FLS:

$$y(\mathbf{x}^{(t)}) = \frac{y_l(\mathbf{x}^{(t)}) + y_r(\mathbf{x}^{(t)})}{2} \quad (12)$$

Step6 Determined explicit dependence for membership functions y_l and y_r .

Step7 Set $t=t+1$, when $t=N+1$, turn to Step 8; otherwise, turn to Step4;

Step8 Set $count=count+1$, if $count=E$, the training process is over, otherwise turn to Step3.

5. Illustration

CSTR is a continuous stirred reactor for polymerization, and its structure is shown in Fig. 1. Considering that CSTR is an irreversible exothermic process of AB with single input and single output, and whether the product concentration meets the requirements largely depends on the reaction temperature. Therefore, in order to ensure the normal reaction rate and meet the product concentration requirements, it is very important to establish an accurate model between the coolant flow rate and the product concentration.

The dynamic characteristic equation of CSTR is as follows:

$$\frac{dC_B}{dt} = \frac{q}{V} (C_A - C_B) - K_0 C_B \exp\left(\frac{-E}{RT}\right) \quad (13)$$

$$\frac{dT}{dt} = \frac{q}{V} (T_f - T) + \frac{(-\Delta H) K_0 C_B}{\rho C_p} \exp\left(\frac{-E}{RT}\right) + \frac{\rho_c C_{pc}}{\rho C_p V} q_c \left[1 - \exp\left(-\frac{hA}{q_c \rho C_{pc}}\right) \right] \times (T_{cf} - T) \quad (14)$$

where, C_B is equilibrium concentration, T is reaction temperature; C_A is feed concentration; in CSTR, the amount of reactant A and product B is equal (amount of catalyst is ignored), q is the feed flow rate; T_f is the feed

temperature; T_{cf} is coolant inlet temperature; the coolant flow q_c is operating variable; $K_0, -\Delta H, \rho_b, \rho_{dc}, C_P, C_{PC}, E/R, V$ chemical reaction coefficients are constant, and the values of CSTR parameters are shown in [16].

When the product concentration is $C_B = 8.36 \times 10^{-2} \text{ mol/L}$, the reaction reaches equilibrium, the CSTR temperature is $T = 440.2 \text{ K}$ and the coolant flow rate is $q_c = 103.4 \text{ L/min}$. At this time, the relationship model between q_c and C_B is established, that is, the steady-state value of coolant flow is superimposed with the limiting white noise in the interval $[-0.002, 0.002]$, which is used as the input to stimulate nonlinear process, and corresponding 4000 groups of data sets of product concentration C_B and reactor output temperature T are generated, as shown in Fig. 2. The first 2000 groups of data is the training data set, and the remaining 2000 groups of data is the test data set.

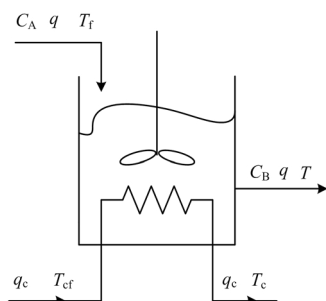


Fig. 1. Structure of CSTR

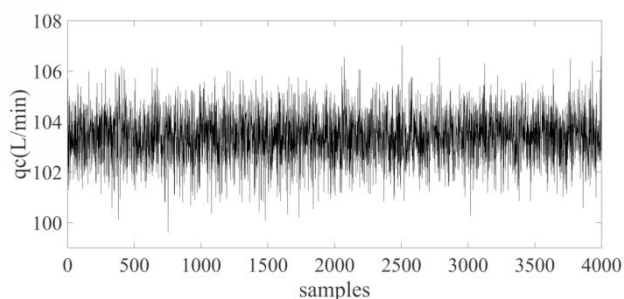


Fig. 2. Input coolant flow q_c to CSTR

Using regression modeling method, the CSTR identification model structure of the system is set as follows:

$$\hat{y}_{C_B}(k) = \Psi(q_c(k-1), q_c(k-2), q_c(k-3), C_B(k-1), C_B(k-2), C_B(k-3)) \quad (15)$$

In the experiment, in different FLS methods, the principal component $p = 4$, each antecedent with 5 fuzzy sets, and the learning rate is 0.4. After 30 cycles of training and testing, MSE is calculated based on test set after each cycle. The selection of initial values of parameters for all different types of FLS is summarized in Table 1, where, σ_x is the standard deviation of training sample data.

Table 1. The parameters and number of parameters in 5 different FLSs.

FLS	Input	For each antecedent (the mean and standard deviation)	Consequent
Type-1 SFLS		$m_k^i, \sigma_k^i = \sigma_x$	$\bar{y}^i \in [0, 0.3]$
Type-1 NSFLS	$\sigma_{X_k} = \sigma_n$	$m_k^i, \sigma_k^i = \sigma_x$	$\bar{y}^i \in [0, 0.3]$
Type-2 SFLS		$[m_k^i - 0.2\sigma_n, m_k^i + 0.2\sigma_n], \sigma_k^i = \sigma_x$	$\bar{y}^i - 0.2\sigma_n, \bar{y}^i + 0.2\sigma_n$
Type-2 NSFLS-1	$\sigma_{X_k} = \sigma_n$	$[m_k^i - 0.2\sigma_n, m_k^i + 0.2\sigma_n], \sigma_k^i = \sigma_x$	$\bar{y}^i - 0.2\sigma_n, \bar{y}^i + 0.2\sigma_n$
Type-2 NSFLS-2	$\sigma_{k1} = \hat{\sigma}_{t10dB}, \sigma_{k2} = \hat{\sigma}_{t0dB}$	$[m_k^i - 0.2\sigma_n, m_k^i + 0.2\sigma_n], \sigma_k^i = \sigma_x$	$\bar{y}^i - 0.2\sigma_n, \bar{y}^i + 0.2\sigma_n$

Figure 3 shows the comparison results between the output of identification model and the actual output value of product concentration after 30 cycles of training based on type-2 nsfls-2 method. The solid line in the figure is the actual value of CSTR, and the dotted line is the output value of type-2 nsfls-2. Figure 4 shows the test error of each point. The results of Fig. 3 and Fig. 4 show that the type-2 nsfls-2 method has good identification effect.

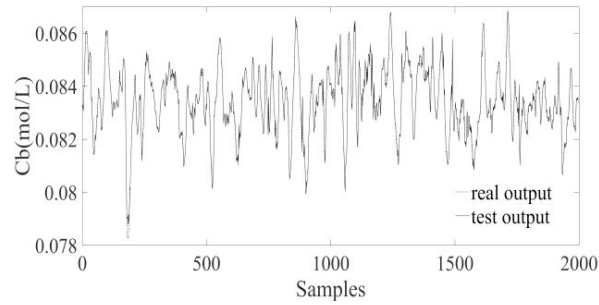


Fig. 3. CSTR identification curve based on type-2 nsfls-2 method

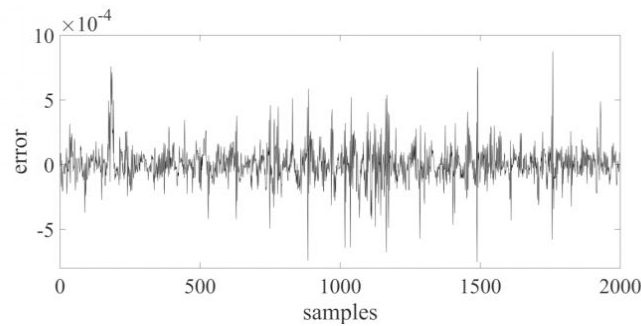


Fig. 4. The error curve on the test set

Table 2 shows the numerical comparison of five different FLS methods on the test set, and the comparison of identification results based on the standard SVM [8] method is also given in the table. It can be seen from Table 2 that, after several cycles of parameter adjustment, the identification accuracy of different FLS methods is obviously not lower than that of standard SVM method. Among them, the identification accuracy of interval type-2 FLS method is significantly higher than that of type-1 FLS method. From the comparison of three interval type-2 FLS methods, type-2 is better than that of type-1 FLS. The identification accuracy of Type-2 NSFLS-2 method in CSTR chemical process identification is the highest, which reflects the application potential of interval type-2 FLS method in the field of nonlinear system identification.

Table 2. Performance comparison of CSTR process identification

Method	MSE
LS-SVM ^[8]	-1.6e-03
Type-1 SFLS	8.7062e-04
Type-1 NSFLS	6.2546e-04
Type-2 SFLS	4.0229e-08
Type-2 NSFLS-1	2.8937e-08
Type-2 NSFLS-2	2.1007e-08

6. Conclusion

As a new fuzzy logic modeling method, interval type-2 FLS is based on uncertainty rules, which can deal with the potential uncertainty in modeling and minimize the impact of uncertainty. For strongly nonlinear CSTR process

with unknown model, a CSTR identification modeling method based on PCA interval type-2 FLS is proposed. Compared with type-1 FLS and SVM method, the results show that interval type-2 FLS combined with PCA is better than that of type-1 FLS and SVM. The interval type-2 FLS method combined with PCA improves identification accuracy effectively and has better performance, which indicates that interval type-2 FLS method combined with PCA has certain application potential in the field of nonlinear system identification.

References

1. Ding X Q, Yang X L, Yang H. Design of nonlinear CSTR process predictive controller . *Control engineering* 2009, **16**: 145-147.
2. Li D J. Adaptive neural network control of continuous stirred tank reactor . *Acta chem* 2013, **64**: 4674-4680.
3. Guan X C, Liu H, Ma L N, Zhao D Y. Discrete model identification and sliding mode control of continuous stirred tank reactor . *Control engineering* 2019, **26**: 440-447.
4. Beyhan S, Alci M. Mixed structured RBF network for direct inverse control of nonlinear systems. *Soft Computing, Computing with Words and Perceptions in System Analysis, Decision and Control, 2009 Fifth IEEE International Conference, Famagusta, IEEE* 2009: 1-4.
5. Chen F, Jing Z L, Yao X D. A fast parameter learning algorithm for fuzzy neural networks . *Control theory and application* 2002, **19**: 583-587.
6. Xu X M, Zhao D Y, Cao L. PID control tuning based on RBF neural network . *Scientific and technological innovation and application* 2019, **13**: 32-33.
7. Li X H, Li J. Identification of continuous stirred tank reactor based on echo state network. *Information and control*, 2014, **43**: 223 -228.
8. Li H S, Zhong Z Y, Zhang Y L. Predictive control based on least squares support vector machine. *Computational technology and automation* 2009, **28**: 32—36.
9. Mendel J M, *Uncertain rule-based fuzzy system: introduction and new directions*. New York: Prentice-Hall, 2000.
10. Mendel J M. General type-2 fuzzy logic systems made simple: A Tutorial. *IEEE Transactions on Fuzzy Systems* 2014, **22**: 1162-1182.
11. Chen Y, Wang D, Tong S. Forecasting studies by designing Mamdani interval type-2 fuzzy logic systems: With the combination of BP algorithms and KM algorithms. *Neurocomputing* 2016, **174**: 1133-1146.
12. Dorantes P N M, Mendez G M. Type-2 fuzzy logic systems for temperature evaluation in ladle furnace. *IEEE Latin America Transactions* 2016, **14**: 3914-3920.
13. Bi Y R; Lu X B, Sun Z, et al. Optimal type-2 fuzzy system for arterial traffic signal control. *IEEE Transactions on Intelligent Transportation Systems* 2018, **19**: 3009-3027.
14. Zhu Q X, Zhang X H, Gu X B, et al. Functional connectivity neural network based on feature extraction and its application in chemical process modeling. *Acta chem* 2018, **69**: 907-912.
15. Mendel J M, John R I, Liu Feilong. Interval type- 2 fuzzy logic systems made simple. *IEEE Transactions on Fuzzy Systems* 2006, **14**: 808—821.
16. Liu Y, Wang H, Yu J, et al. Selective recursive kernel learning for online identification of nonlinear systems with NARX form. *Journal of Process Control* 2010, **20**: 181-194.